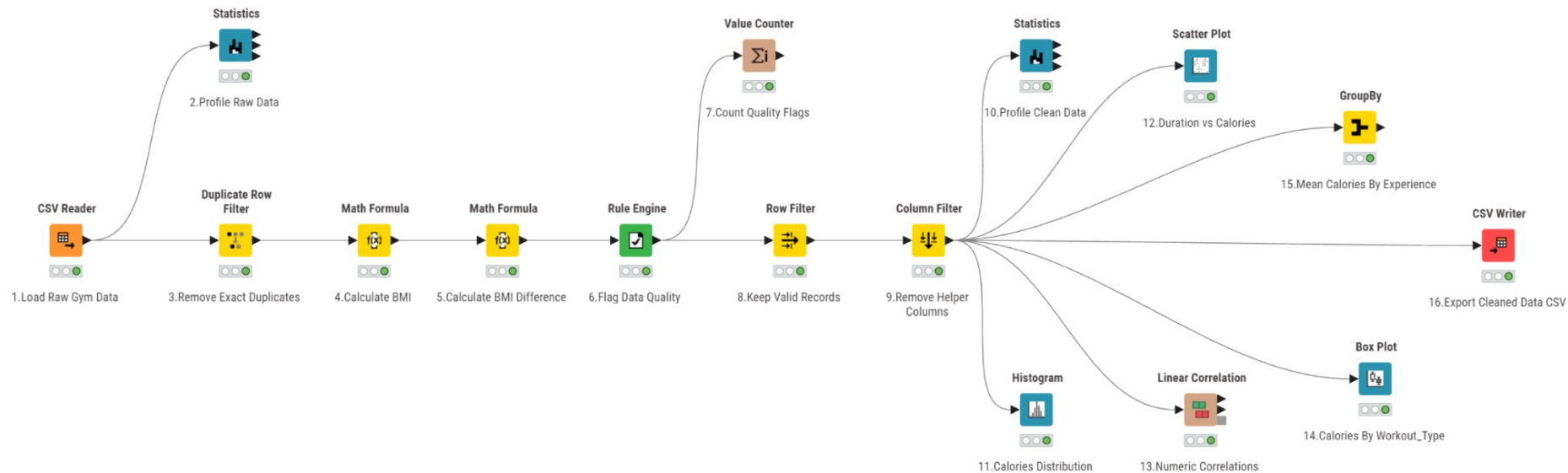


Wai Yan Hpone Lat
Admin No: 253955B



Executed workflow - green node status confirms successful processing

Profile first

Statistics checked ranges, distributions and missing values before any row was removed.

Validate each row

Duplicates, BMI consistency and heart-rate logic were checked so modelling would not learn from invalid records.

Export only clean fields

Temporary quality columns were removed before the final 15-variable CSV was passed to SAS Viya.

KNIME was used for preparation and exploration only. All predictive models and model evaluation were completed in SAS Viya.

Profiling exposed the main data-quality risk

► 1: File Table Flow Variables

Rows: 4865 | Columns: 15

Table Statistics

#	RowID	Age 100 Number (L)	Gender T String	Weight (kg) 100 Number (L)	Height (m) 100 Number (L)	Max_BPM 100 Number (L)	Avg_BPM 100 Number (L)	Resting_... 100 Number (L)	Session_... 100 Number (L)	Calories_... 100 Number (L)	Workout_... T String	Fat_Perce... 100 Number (L)	Wat 100 N
1	Row0	56	Male	88.3	1.71	180	157	60	1.69	1313	Yoga	12.6	3.5
2	Row1	46	Female	74.9	1.53	179	151	66	1.3	883	HIIT	33.9	2.1
3	Row2	32	Female	68.1	1.66	167	122	54	1.11	677	Cardio	33.4	2.3
4	Row3	25	Male	53.2	1.7	190	164	56	0.59	532	Strength	28.8	2.1
5	Row4	38	Male	46.1	1.79	188	158	68	0.64	556	Strength	29.2	2.8

Raw table: 4,865 observations x 15 variables

4,865

raw observations

15

variables

0

missing values

3,892

exact duplicate rows

► 1: Statistics Table ► 2: Nominal Histogram Table ► 3: Occurrences Table Flow Variables

Rows: 13 | Columns: 16

Table Statistics

Column T String	Min 100 Number (L)	Max 100 Number (L)	Mean 100 Number (L)	Std. devia... 100 Number (L)	Variance 100 Number (L)	Skewness 100 Number (L)	Kurtosis 100 Number (L)	Overall su... 100 Number (L)
Age	18	59	38.683	12.176	148.253	-0.078	-1.215	188,195
Weight (kg)	40	129.9	73.855	21.199	449.388	0.771	-0.029	359,303
Height (m)	1.5	2	1.723	0.128	0.016	0.338	-0.726	8,380.35
Max_BPM	160	199	179.884	11.521	132.732	-0.038	-1.188	875,135
Avg_BPM	120	169	143.767	14.339	205.613	0.086	-1.199	699,425
Resting_BPM	50	74	62.223	7.324	53.642	-0.072	-1.182	302,715
Session_Duratio	0.5	2	1.256	0.343	0.118	0.026	-0.354	6,112.5
Calories_Burner	303	1,783	905.422	272.529	74,272.267	0.278	-0.061	4,404,880
Fat_Percentage	10	35	24.977	6.257	39.148	-0.634	-0.343	121,512
Water_Intake (li	1.5	3.7	2.627	0.6	0.36	0.071	-1.021	12,778.5
Workout_Frequ	2	5	3.322	0.913	0.833	0.15	-0.808	16,160
Experience_Lev	1	3	1.81	0.739	0.547	0.318	-1.123	8,805
BMI	12.32	49.84	24.912	6.658	44.331	0.763	0.735	121,197.5

Statistics node: target mean 905.42 kcal; range 303 to 1,783

Why this mattered

- Duplicates made up about 80% of the raw rows.
- Keeping them would over-represent repeated observations and make evaluation look more reliable than it really was.

Rows: 16 | Columns: 14

Table Statistics

Name	Type	# Missing val...	# Unique valu...	Minimum	25% Quantile	50% Quantile	75% Quantile	Mean	Mean Absolu...	Standard Dev...	Sum
Column	String	0	13	0	0	0	0	0	0	0	0
Min	Number (Float)	0	12	0.5	303	1.5	12.32	85	55.371	64.137	85.735
Max	Number (Float)	0	12	2	1,783	3.35	49.84	149.45	193.418	245.41	482.225
Mean	Number (Float)	0	13	1.256	905.422	2.218	24.977	108.811	112.651	137.095	244.938
Std. deviation	Number (Float)	0	13	0.128	272.529	0.67	6.658	13.258	27.287	37.73	73.975
Variance	Number (Float)	0	13	0.016	74,272.267	0.453	44.331	176.933	5,795.942	10,534.819	20,574.968
Skewness	Number (Float)	0	13	-0.634	0.771	-0.055	0.086	0.328	0.152	0.263	0.367
Kurtosis	Number (Float)	0	13	-1.215	0.735	-1.185	-0.808	-0.202	-0.655	0.496	0.607
Overall sum	Number (Float)	0	13	4,404.880	10,791.75	121,512	529,364	546,046.065	666,969.508	1,191,624.66	7,124.5
No. missing	Number (Integer)	0	1	0	0	0	0	0	0	0	0
No. NaNs	Number (Integer)	0	1	0	0	0	0	0	0	0	0
No. +oos	Number (Integer)	0	1	0	0	0	0	0	0	0	0
No. -oos	Number (Integer)	0	1	0	0	0	0	0	0	0	0
Median	Number (Float)	13	0	0	0	0	0	0	0	0	0
Row count	Number (Integer)	0	1	4,865	4,865	4,865	4,865	4,865	0	0	63,245
Histogram	Image (SVG)	0	13	0	0	0	0	0	0	0	0

- Profiling also identified records where Avg_BPM was greater than Max_BPM.

The main issue was not missing data; it was extensive exact duplication plus 23 logically invalid heart-rate records.

Cleaning reduced 4,865 raw rows to 950 valid rows

4,865

raw rows

deduplicate

973

after 3,892 duplicates removed

validate

950

after 23 invalid BPM rows removed

► 1: Filtered/Labeled Data 🔍 Flow Variables

Rows: 973 | Columns: 15

Table 📊 Statistics 📈

#	RowID	Age Number (L...)	Gender T: String	Weight (kg) Number (L...)	Height (m) Number (L...)	Max_BPM Number (L...)	Avg_BPM Number (L...)	Resting... Number (L...)	Session... Number (L...)	Calories... Number (L...)	Workout... T: String	Fat_Perce... Number (L...)	Wa... I
1	Row0	56	Male	88.3	1.71	180	157	60	1.69	1313	Yoga	12.6	3.5
2	Row1	46	Female	74.9	1.53	179	151	66	1.3	883	HIIT	33.9	2.1
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4	Row3	25	Male	53.2	1.7	190	164	56	0.59	532	Strength	28.8	2.1
5	Row4	38	Male	46.1	1.79	188	158	68	0.64	556	Strength	29.2	2.8

Duplicate Row Filter output - 973 unique records

► 1: Included Rows 🔍 Flow Variables

Rows: 950 | Columns: 18

Table 📊 Statistics 📈

☐	#	RowID	Age Number (L...)	Gender T: String	Weight (kg) Number (L...)	Height (m) Number (L...)	Max_BPM Number (L...)	Avg_BPM Number (L...)	Resting... Number (L...)	Session... Number (L...)	Calories... Number (L...)	Workout... T: String	Fat_Perce... Number (L...)	Wa... I
☐	1	Row0	56	Male	88.3	1.71	180	157	60	1.69	1313	Yoga	12.6	3.
☐	2	Row1	46	Female	74.9	1.53	179	151	66	1.3	883	HIIT	33.9	2.
☐	3	Row2	32	Female	68.1	1.66	167	122	54	1.11	677	Cardio	33.4	2.
☐	4	Row3	25	Male	53.2	1.7	190	164	56	0.59	532	Strength	28.8	2.
☐	5	Row4	38	Male	46.1	1.79	188	158	68	0.64	556	Strength	29.2	2.

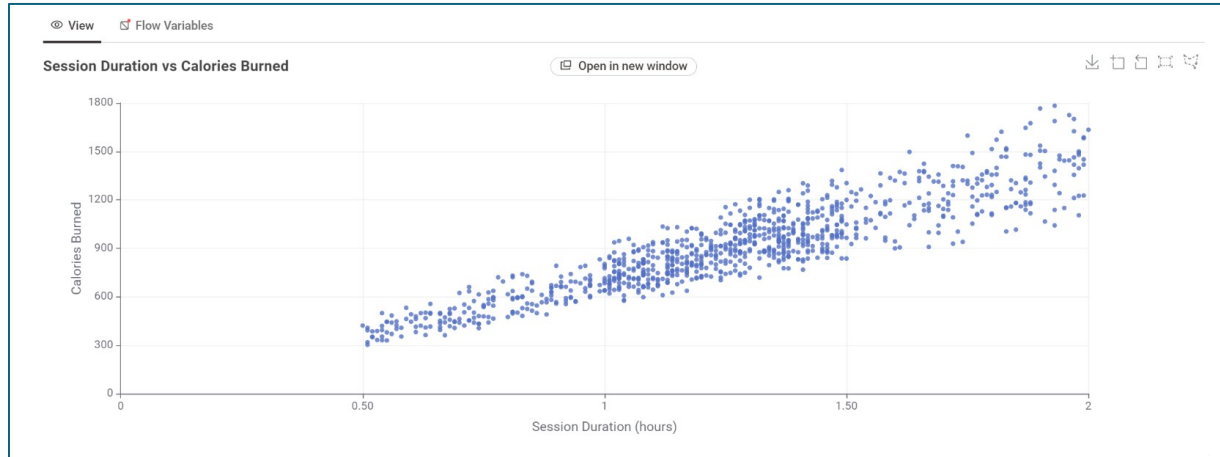
Row Filter output - 950 valid rows before helper-column removal

Cleaning decisions

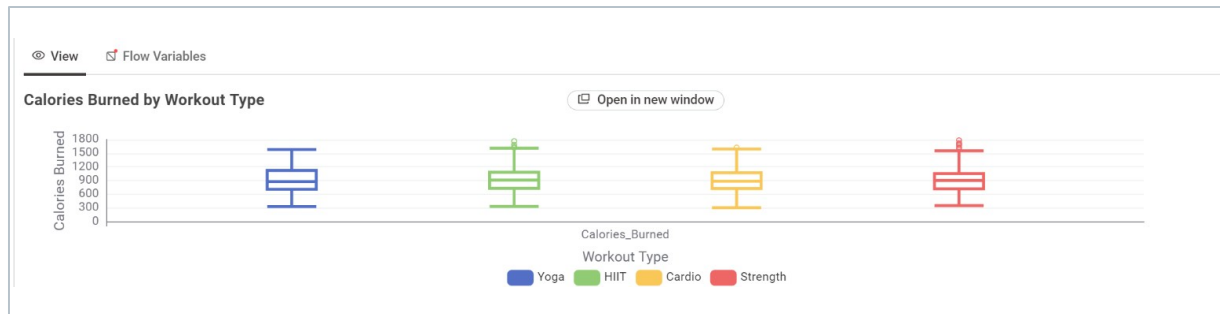
- Recalculated BMI from weight and height to check the supplied BMI.
- Applied the rule $\text{Max_BPM} \geq \text{Avg_BPM}$; 23 failures were removed because the recorded maximum cannot be below the average.
- Removed Data_Quality_Flag, BMI_Calculated and other helper fields before export.
- Retained plausible extreme calorie values because they may represent genuine high-intensity sessions.

Final modelling file: 950 observations x 15 variables, with 0 missing values, 0 duplicates and 0 invalid heart-rate records.

Session duration had the strongest relationship with calories



Session duration vs Calories_Burned



Calories_Burned by Workout_Type

1: Correlation measure 2: Correlation matrix 3: Correlation model 4: Flow Variables												
Rows: 13 Columns: 13												
RowID	Age	Weight (kg)	Height (m)	Max_BPM	Avg_BPM	Resting_BPM	Session_Duration (hours)	Calories_Burned	Fat_Percentage	Water_Intake (liters)	Workout_Frequency	Experience_Level
1		-0.041	-0.025	-0.012	0.032	0.008	-0.017	-0.154	0	0.045	0.009	-0.018
Weight	-0.041	1	0.37	0.058	0.014	-0.033	-0.013	0.097	-0.228	0.398	-0.011	0.006
Height	-0.025	0.37	1	-0.03	-0.004	0.001	-0.004	0.097	-0.243	0.397	-0.008	-0.002
Max_B	-0.012	0.058	-0.03	1	0.017	0.038	-0.004	0.005	-0.003	0.021	-0.045	-0.012
Avg_BF	0.032	0.014	-0.004	0.017	1	0.062	0.031	0.35	-0.013	0.01	0.005	0.013
Resting	0.008	-0.033	0.001	0.038	0.062	1	-0.023	0.01	-0.011	0.011	-0.013	-0.005
Session	-0.017	-0.013	-0.004	-0.004	0.031	-0.023	1	0.909	-0.584	0.289	0.646	0.764
Calorie	-0.154	0.097	0.097	0.005	0.35	0.01	0.909	1	-0.601	0.366	0.581	0.696
Fat_Pe	0	-0.228	-0.243	-0.003	-0.013	-0.011	-0.584	-0.601	1	-0.592	-0.54	-0.657
Water	0.045	0.398	0.397	0.021	0.01	0.011	0.289	0.366	-0.592	1	0.241	0.307
Workout	0.009	-0.011	-0.008	-0.045	0.005	-0.013	0.646	0.581	-0.54	0.241	1	0.838
Experie	-0.018	0.006	-0.002	-0.012	0.013	-0.005	0.764	0.696	-0.657	0.307	0.838	1
BMI	-0.02	0.853	-0.155	0.076	0.019	-0.036	-0.01	0.055	-0.118	0.217	0.002	0.014

Pearson correlations from the 950 clean rows

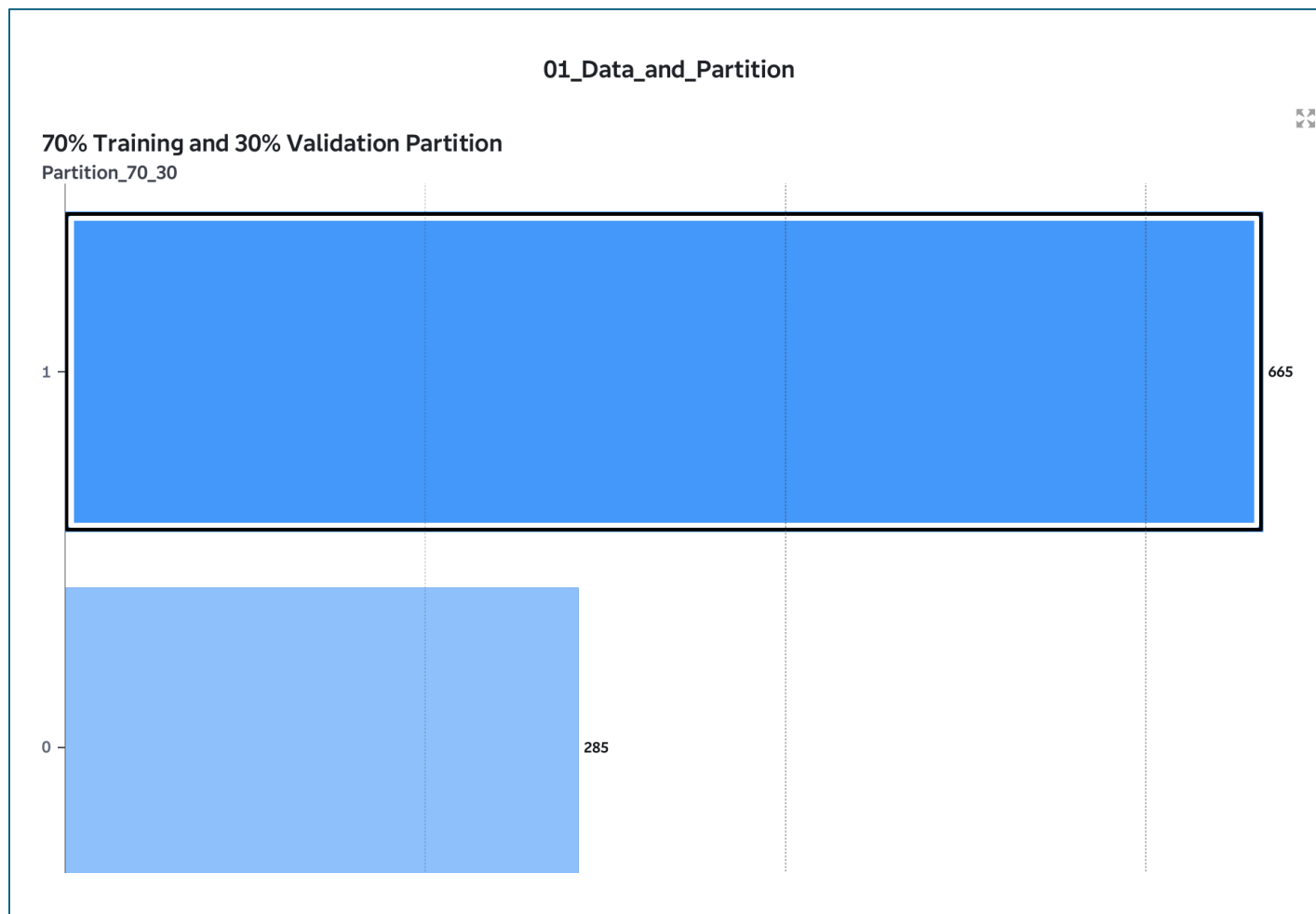
Strongest relationships with Calories_Burned

Session_Duration (hours)	$r = 0.909$
Experience_Level	$r = 0.696$
Fat_Percentage	$r = -0.601$
Workout_Frequency	$r = 0.581$
Water_Intake (liters)	$r = 0.366$
Avg_BPM	$r = 0.350$

The scatter plot rises clearly, while the workout-type box plots overlap. This suggests that duration and intensity contain more useful predictive information than workout category alone.

Session duration was the strongest single signal. The widening spread suggested that a flexible model might capture patterns beyond a simple straight line.

SAS Viya roles and one shared partition made the test fair



SAS partition evidence: value 1 = training; value 0 = validation

665

training records (70%)

285

validation records (30%)

Fair evaluation design

- Imported the KNIME output: 950 cleaned observations and 15 modelling variables.
- Assigned Calories_Burned as the continuous response; the remaining 14 variables were candidate predictors.
- Data Partition created 665 training rows (70%) and 285 validation rows (30%). The same partition was used for all four models.
- Validation ASE was the main selection criterion. Lower ASE means lower squared prediction error on unseen validation observations.

Selection rule: choose the lowest Validation ASE

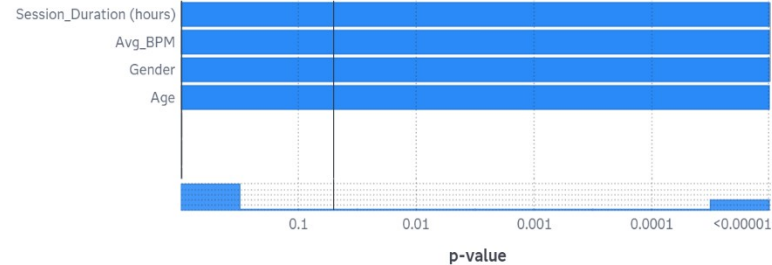
Using one shared partition made the comparison fair: every model learned from the same 665 rows and was checked on the same 285 unseen rows.

Linear Regression performed well, but residuals still curved

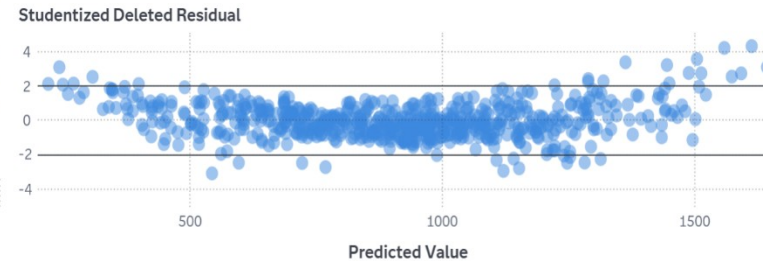
Linear Regression of Calories_Burned

Fit: Validation ASE 1.5K ▾ Observations: 950 of 950

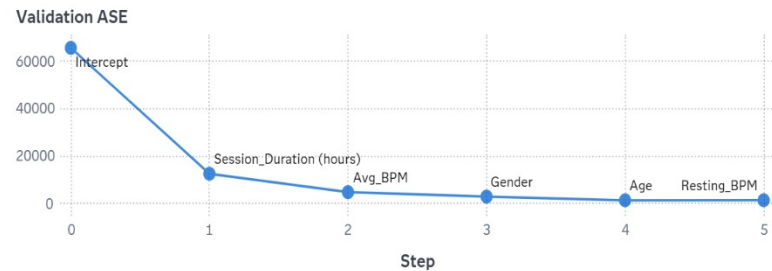
Fit Summary



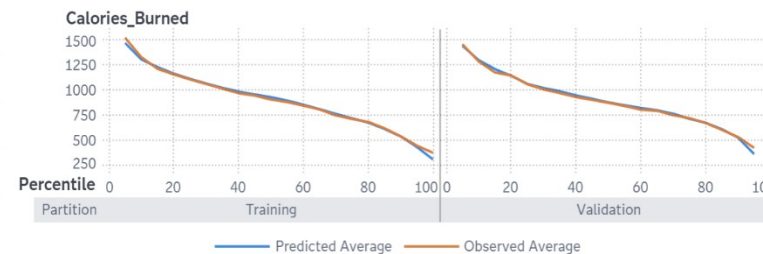
Residual Plot



Variable Selection Plot



Assessment



Actual SAS Linear Regression fit summary and residual diagnostic

The model was accurate enough to be a useful benchmark, but the residual shape showed a systematic pattern that a straight-line specification did not capture.

1,588.9673

Training ASE

1,469.6926

Validation ASE

38.34 kcal

Validation RMSE

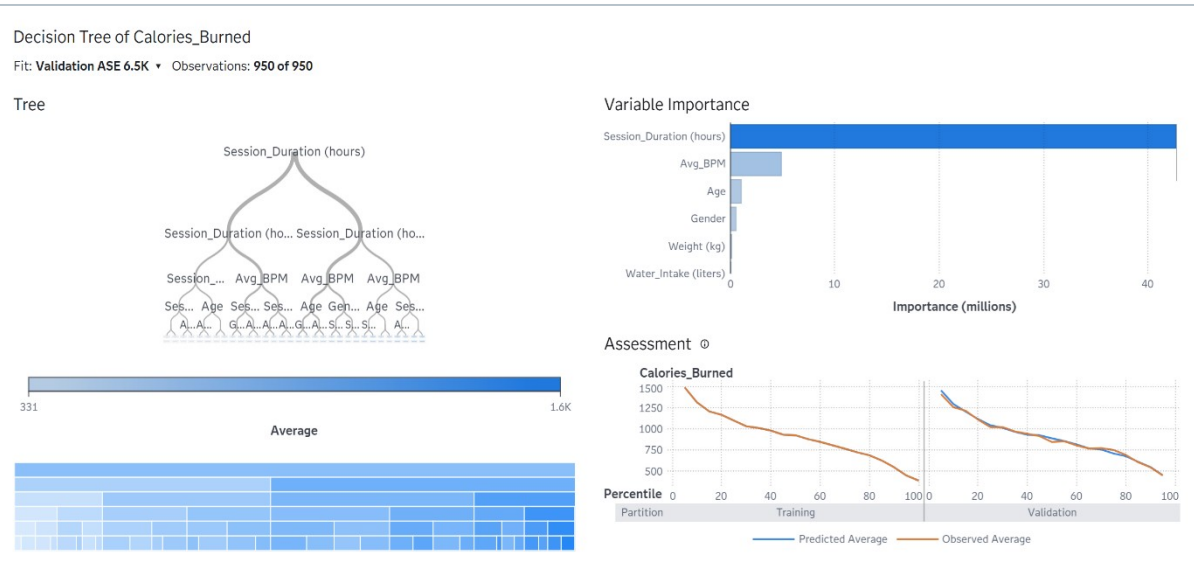
Diagnostic interpretation

- Predictive performance was relatively strong.
- Training and validation errors were reasonably similar, so there was no strong evidence of serious overfitting.
- The visible curved, U-shaped residual pattern suggests that the predictor-target relationship was not completely linear.
- This supported testing nonlinear tree-based and ensemble models.

Linear Regression became the second-best model. Its similar training and validation ASEs were reassuring, but the residual curvature justified a nonlinear alternative.

The tree-based alternatives generalised less well

Decision Tree



3,996.1021

Training ASE

6,457.5496

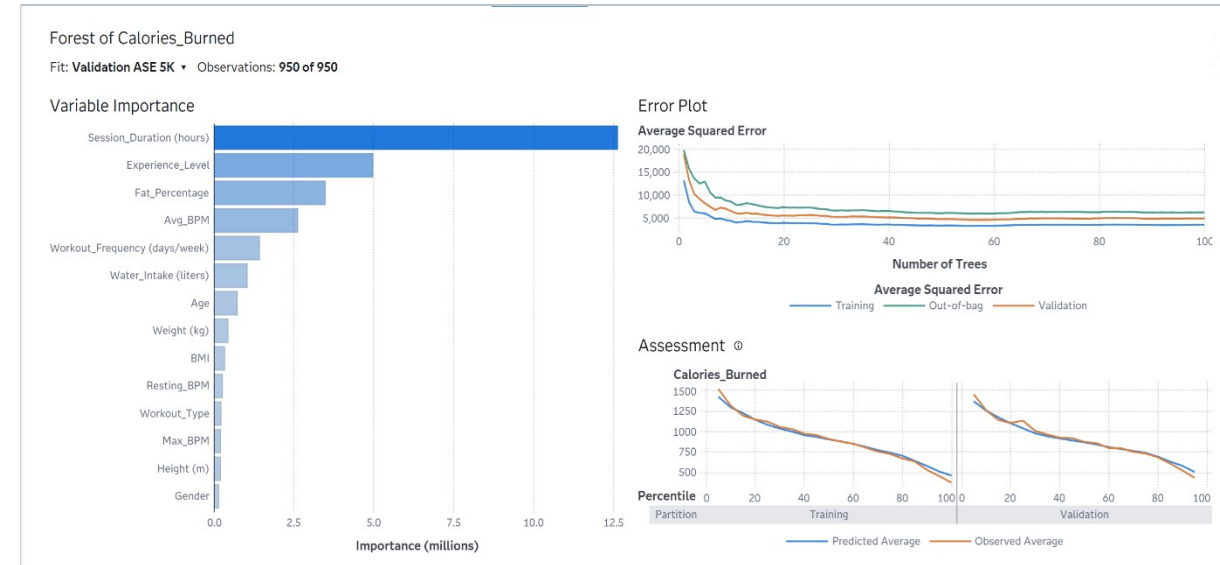
Validation ASE

80.36

Validation RMSE (kcal)

Validation error was substantially higher than training error, so the tree overfit and generalised poorly compared with Linear Regression. Session_Duration (hours) was dominant and drove the first split in the SAS evidence.

Forest



3,591.2230

Training ASE

4,982.7305

Validation ASE

70.59

Validation RMSE (kcal)

Combining many trees improved on the single Decision Tree and reduced some instability. The visible error curve fell and then levelled off, yet Validation ASE remained far above Linear Regression and Gradient Boosting.

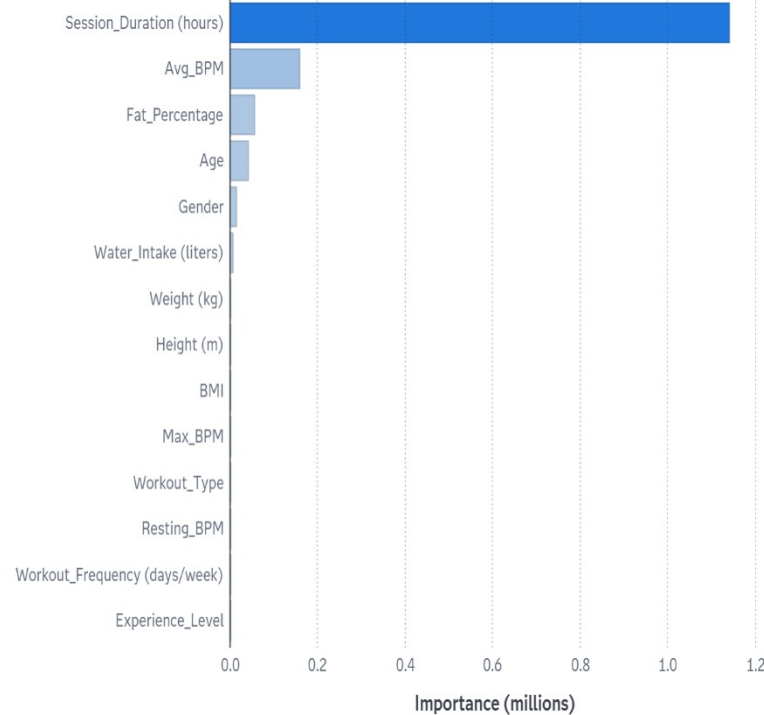
Forest was better than the single tree, but neither tree-based alternative matched the validation accuracy of Linear Regression or Gradient Boosting.

Gradient Boosting achieved the lowest validation error

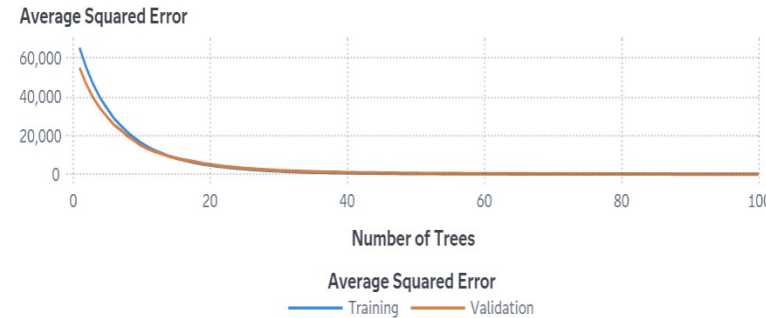
Gradient Boosting of Calories_Burned

Fit: Validation ASE 432 ▾ Observations: 950 of 950

Variable Importance



Iteration Plot



Assessment @



234.3717

Training ASE

431.8658

Validation ASE

20.78 kcal

Validation RMSE

Why it became champion

- It had the lowest Validation ASE of all four models, substantially below Linear Regression, Decision Tree and Forest.
- A training-validation gap remained, so some overfitting may still be present.
- Selection was based on performance on the 285 unseen validation observations, not on training fit alone.

Important predictive signals: **Session_Duration (hours)**, **Avg_BPM** and **Fat_Percentage**. Importance indicates predictive association, not causation.

Actual SAS Gradient Boosting diagnostics and variable importance

The validation result supports Gradient Boosting as the strongest model in this experiment, while the remaining error gap still warrants monitoring on new data.

Model Comparison of Calories_Burned

Model Comparison Calories_Burned

Fit Statistic

Validation ASE



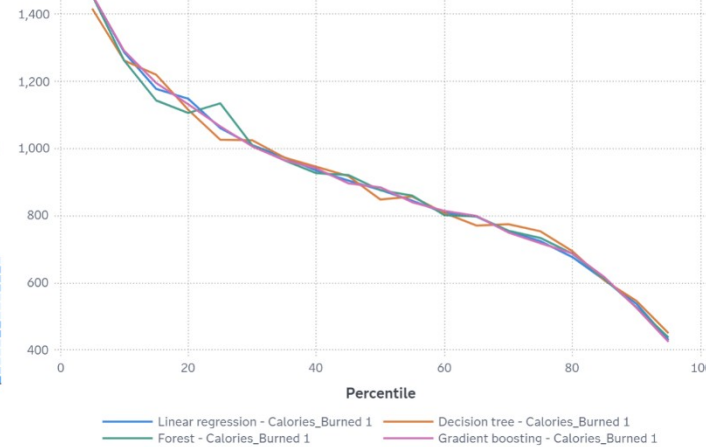
Relative Importance Plot

Variable



Assessment

Validation Observed Average



CHAMPION MODEL

Gradient Boosting

431.8658

Validation ASE

70.6% lower than Linear Regression, the second-best model. All four models used the same 285 validation observations.

Actual SAS Model Comparison - Gradient Boosting marked Selected

Model	Training ASE	Validation ASE	Validation RMSE
Linear Regression	1,588.9673	1,469.6926	38.34
Decision Tree	3,996.1021	6,457.5496	80.36
Forest	3,591.2230	4,982.7305	70.59
Gradient Boosting	234.3717	431.8658	20.78

With one shared partition, the comparison was consistent. Gradient Boosting had the lowest error on the 285 unseen validation observations, which was the main basis for champion selection.

Recommended model

Gradient Boosting

Validation ASE ~ 431.87 | Validation RMSE ~ 20.78 kcal

Best predictive accuracy among the four tested models under the current validation design.

Practical interpretation

- Session Duration was consistently one of the strongest predictive signals.
- Heart-rate and intensity information also contributed.
- Predictions should be presented as estimates, not guaranteed calorie measurements.

Future improvement

- Validate the champion on genuinely new workout data and monitor prediction error after deployment.
- Retain the KNIME duplicate, BMI and heart-rate checks. Retrain when data distributions or performance change.

Limitations

- Only 950 cleaned observations from one dataset were available.
- One 70/30 training-validation partition was used.
- No completely independent external test dataset was available.
- Gradient Boosting still showed a training-validation performance gap.
- Variable importance indicates predictive association, not causation.
- Performance may change for different populations or future workout patterns.

Conclusion

KNIME produced a validated modelling dataset, while SAS Viya provided a fair comparison of four predictive models. Gradient Boosting achieved the lowest validation error and was therefore selected as the recommended model for predicting Calories_Burned, subject to continued monitoring and validation on new data.